

Network Design and Simulation Project

Egyptian Russian University Campus Network

Course: Network Fundamentals
Department: MIS

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Tool Used:	Cisco Packet Tracer
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1. Project Overview and Chosen Scenario

The chosen work environment for this network simulation project is the Egyptian Russian University (ERU). A university campus represents a highly realistic and complex networking scenario, requiring robust infrastructure to serve a large and diverse population of users across physically separate buildings and faculties.

The university network is designed to serve the following distinct departments/zones:

1.1 Department Breakdown

Department / Zone	Description	Key Devices	Subnet
Main Campus (eru.edu.eg)	Central administration and main student labs. Houses the largest concentration of student PCs connected via Switch0.	18 Student PCs (student32–student49), Switch0, Router0	192.168.7.0/24
Faculty of Computer & Info. (fci.eru.edu.eg)	Faculty wing hosting student computing labs. Connected via Switch1 and Router1.	18 Student PCs (student50–student67), Switch1, Router1	192.168.2.0/24
Datacenter / Server Farm	Central datacenter hosting all university servers: DNS, SMTP Mail, DHCP, Web (LMS, Library, Student's portal, ERU main site). Connected via Switch2.	8 Servers, Switch2, Router2 (core router)	192.168.3.0/24
Campus Right Wing (lms.eru.edu.eg)	Additional student labs and faculty offices. Includes printers shared across the wing.	17 Student PCs (student15–student31), 2 Printers, Switch5, Router5	192.168.6.0/24
Library Building (library.eru.edu.eg)	University library zone. Hosts the library server and student terminals for research access, plus printers.	12 Student PCs (student1–student12), 1 Printers, Library Server, Switch3, Router3	192.168.4.0/24
Annex / Extended Lab (lms.eru.edu.eg) (Switch4)	Smaller annex lab connected to the library zone via Router4, with additional student PCs and a printer.	2 Student PCs (student13–student14), 1 Printer, Switch4, Router4	192.168.5.0/24

2. servers

Server Name / Hostname	IP Address	Service Provided
DNS Server	192.168.3.4	Resolves all eru.edu.eg domain names
DHCP Server	192.168.3.3	Automatically assigns IP addresses to all student PCs across all subnets
SMTP Server (eru.edu.eg)	192.168.3.5	Handles outgoing/incoming university email
Server eru.edu.eg	192.168.3.6	ERU main website / portal
Server fci.eru.edu.eg	192.168.3.7	Faculty of Business Informatics web server
Server library.eru.edu.eg	192.168.3.8	Library portal and catalog server
Server students.eru.edu.eg	192.168.3.9	Student information and grade portal
Server lms.eru.edu.eg	192.168.3.10	Learning Management System (LMS) for courses
Library Server (local)	Static (Switch3)	Local library resource server within Library building

2.1 Cable Types

- Straight-through Ethernet cables: Used to connect PCs and printers (end devices) to switches, and switches to routers.
- Cross-over Ethernet cables (or auto-MDI-X): Used for router-to-router connections (WAN links) shown as red lines in the simulation diagram.
- Serial cables (DCE/DTE): Optionally used for WAN links between routers if serial interfaces are configured instead of FastEthernet.

3. Network Topologies

The network design incorporates three distinct topology types as required:

3.1 Star Topology (Within Each Department LAN)

Each campus department (Main Campus, FCI, Library, Right Wing, Annex) is internally structured as a Star Topology. All end devices (PCs and printers) connect via individual cables directly to a central switch. This is the most common LAN topology and offers:

- Easy fault isolation a single device failure does not affect others.
- Simple addition of new devices without disrupting the network.
- Centralized management through the switch. Devices affected: Switch0, Switch1, Switch3, Switch4, Switch5 each acting as the central hub of its star.

3.2 Hierarchical / Partial Mesh Topology (Router Interconnections)

The routers connecting the campus departments form a Hierarchical/Partial Mesh topology with Router2 acting as the core. The connections are:

- Router0 connects to Router1 (Main Campus to FCI link).
- Router1 connects to Router2 (FCI to Core).
- Router2 connects to Router3 (Core to Library).
- Router3 connects to Router4 (Library to Annex).
- Router5 connects back to the core area (Right Wing).

This hierarchical design with a central core router provides scalability, redundancy potential, and efficient routing between all campus segments.

3.3 Star Topology (Datacenter)

The Datacenter itself uses a Star Topology, with all servers connecting to Switch2, which in turn connects to the core Router2. This isolates server traffic and simplifies server management and security.

4. Network Services Implemented

4.1 DHCP Server

A DHCP Server has been configured on the Datacenter subnet (192.168.3.0/24). The server automatically assigns IP addresses to all student PCs across all campus departments, eliminating the need for manual IP configuration.

DHCP Configuration Summary:

Scope / Pool	Address Range	Default Gateway
Main Campus Pool	192.168.7.10 – 192.168.7.254	192.168.1.1 (Router0)
FCI Campus Pool	192.168.2.10 – 192.168.2.254	192.168.2.1 (Router1)
Library Pool	192.168.4.10 – 192.168.4.254	192.168.4.1 (Router3)
Annex Lab Pool	192.168.5.10 – 192.168.5.254	192.168.5.1 (Router4)
Right Wing Pool	192.168.6.10 – 192.168.6.254	192.168.6.1 (Router5)

Note: IP helper-address (DHCP relay) is configured on each router interface facing the student LAN subnets, forwarding DHCP broadcast requests to the centralized DHCP server at 192.168.3.3.

4.2 Mail Server (SMTP)

An SMTP Mail Server has been configured at IP address 192.168.3.5, with the domain eru.edu.eg. User email accounts have been created for students and staff. The server handles both sending and receiving of email messages between any devices within the university network.

Configuration steps performed:

- SMTP service enabled on the server at 192.168.3.5.
- Domain set to: eru.edu.eg
- User accounts created (e.g., student1@eru.edu.eg).
- Email client (Outlook / Packet Tracer email app) configured on student PCs with SMTP server IP and credentials.
- Successful send/receive of email demonstrated between PCs in different subnets.

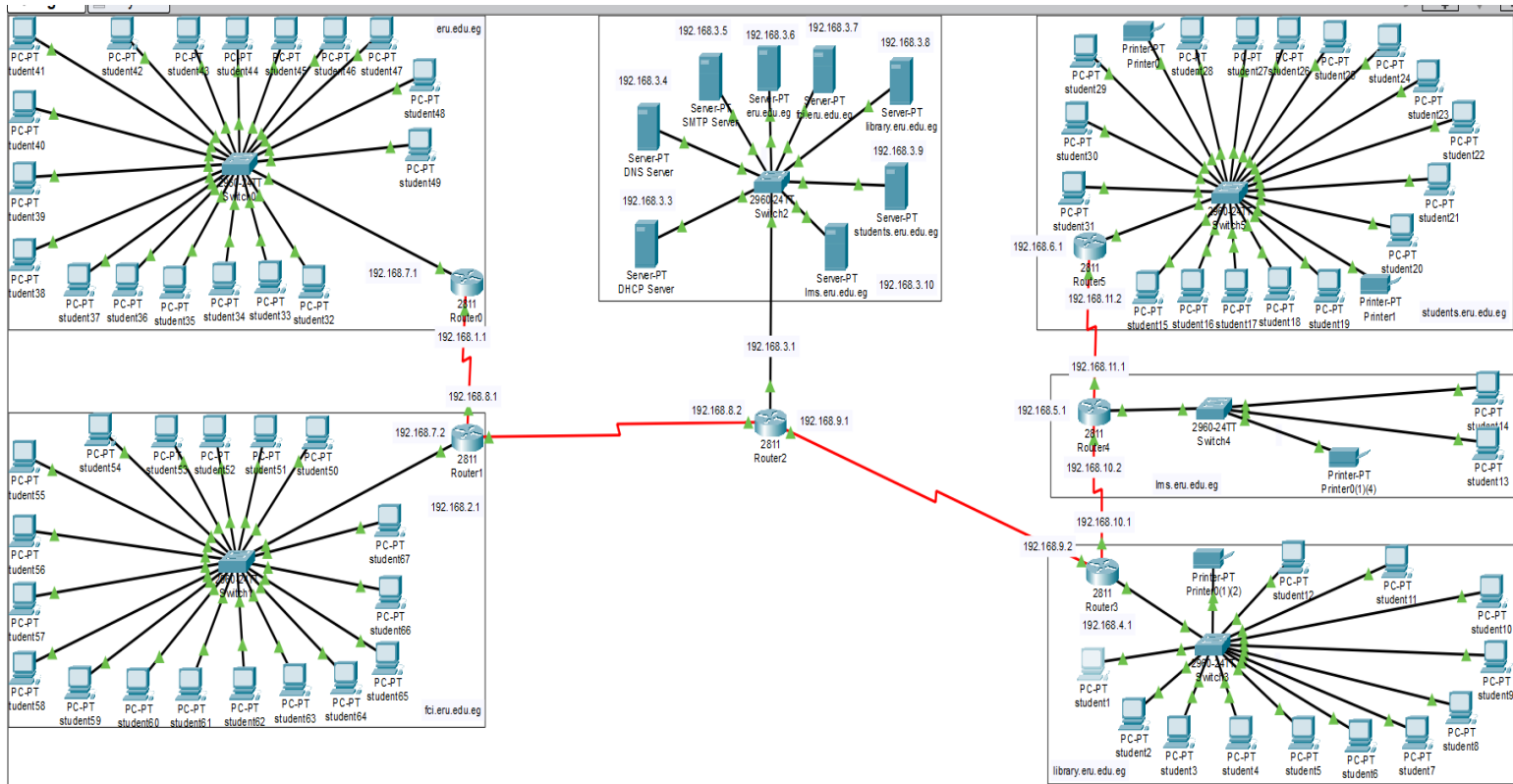
4.3 DNS Server

The DNS Server at 192.168.3.3 resolves all internal university domain names. DNS records configured include:

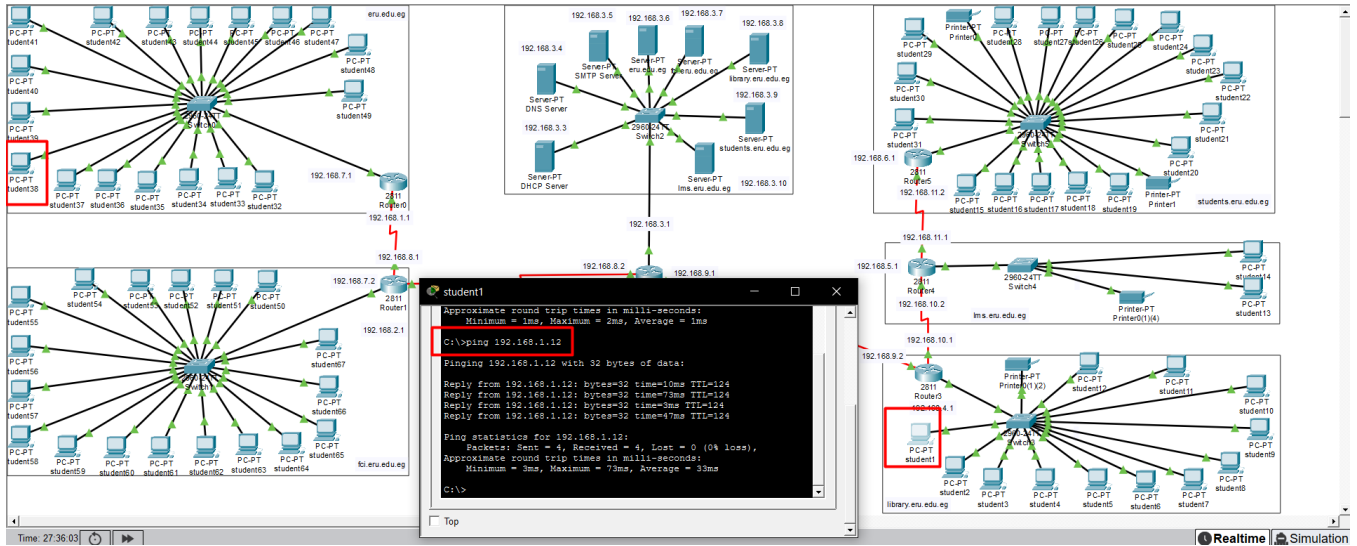
- eru.edu.eg → 192.168.3.6
- fci.eri.edu.eg → 192.168.3.7
- library.eri.edu.eg → 192.168.3.8
- students.eri.edu.eg → 192.168.3.9
- lms.eri.edu.eg → 192.168.3.10
- smtp.eri.edu.eg → 192.168.3.5

5. Verification and Testing

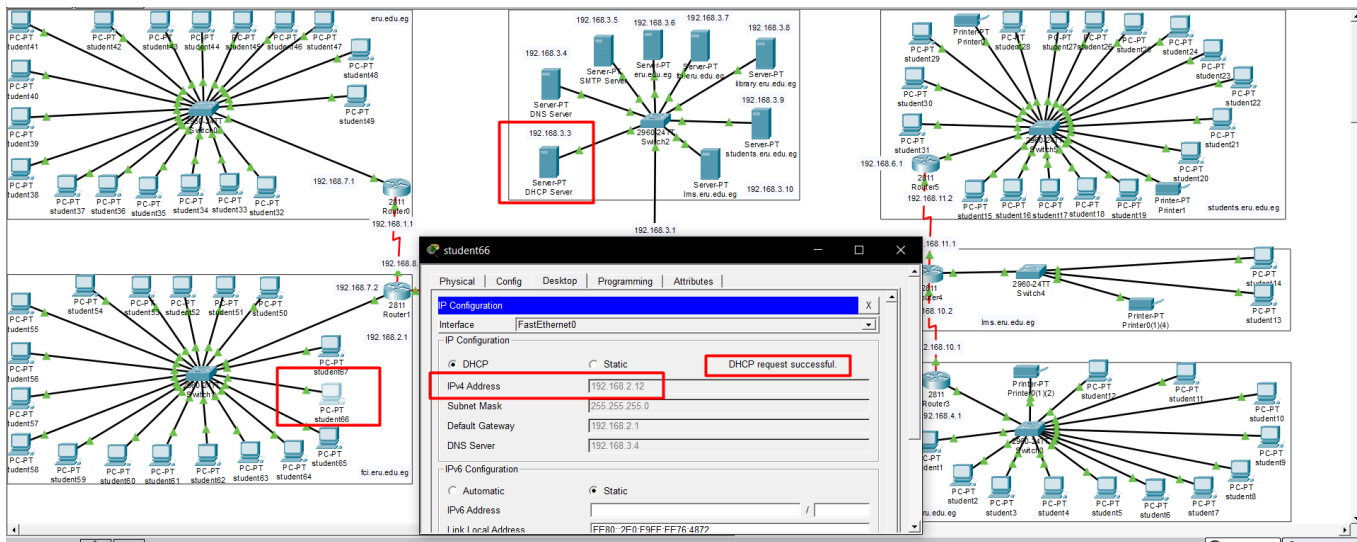
The following tests were performed to verify full network functionality. Screenshots from Cisco Packet Tracer should be attached below each test result.



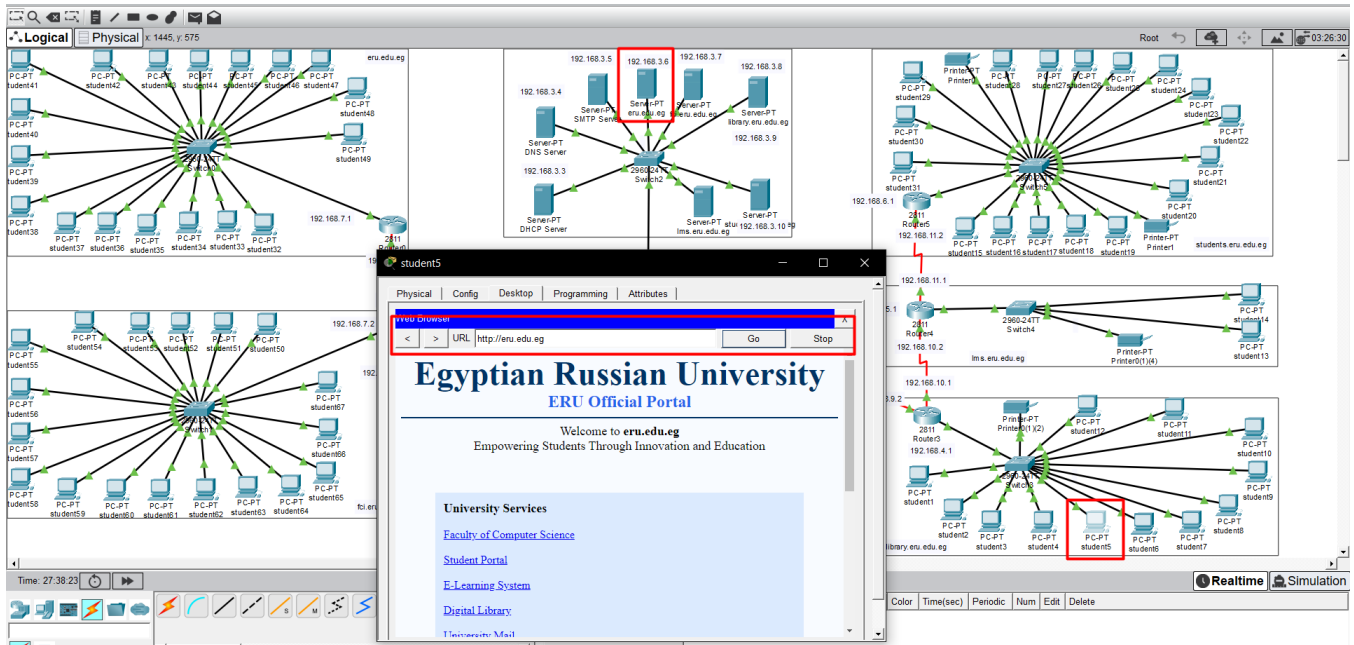
6.1 Ping Tests Between Departments



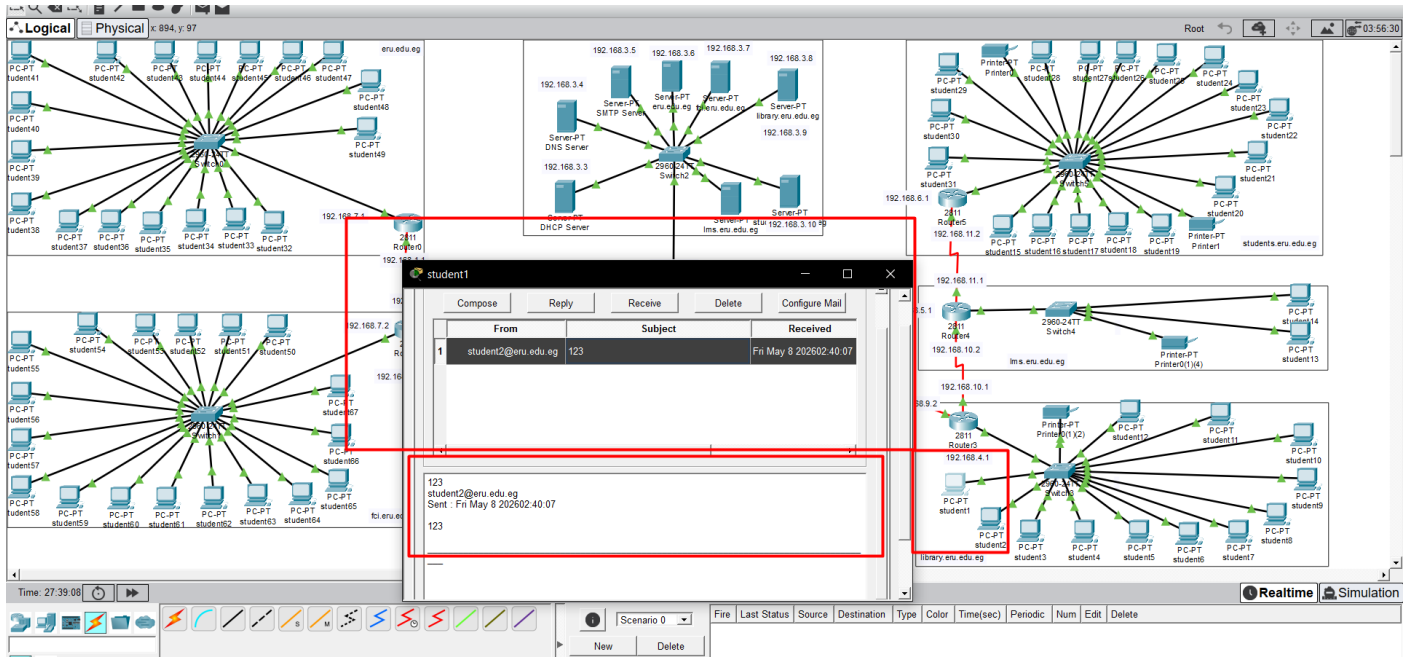
6.2 DHCP Lease Verification



6.3 Web Server Access



6.4 Email Delivery



7. Conclusion

This project successfully demonstrates the design and simulation of a fully functional university campus network for the Egyptian Russian University. The network spans six distinct zones (departments), interconnected through a hierarchical routing architecture with a central core router. All required services DHCP for automatic IP assignment, DNS for name resolution, SMTP for email communication, and HTTP web servers were configured and verified to be operational across all subnets. The network satisfies all project requirements: at least three departments, multiple topology types (Star and Hierarchical Mesh), comprehensive device inventory, and demonstrated network services with verification screenshots.